Software Systems Planning (Gpo 103)

Actividad: Evidence week 1 (Adolfo)

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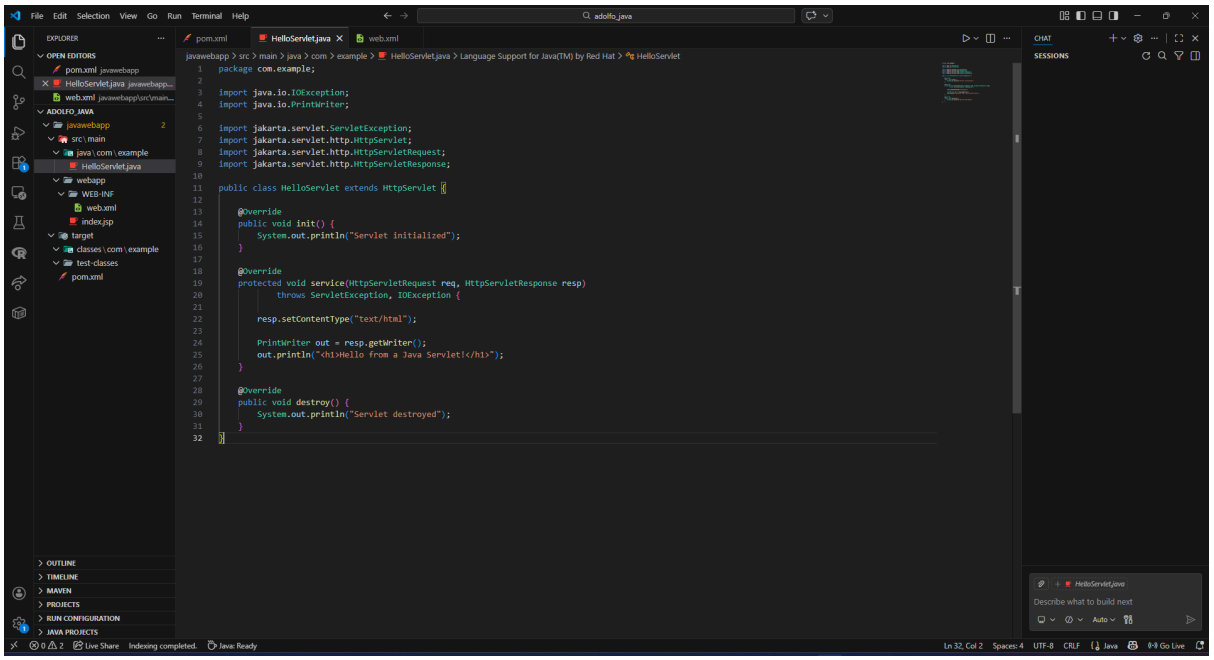
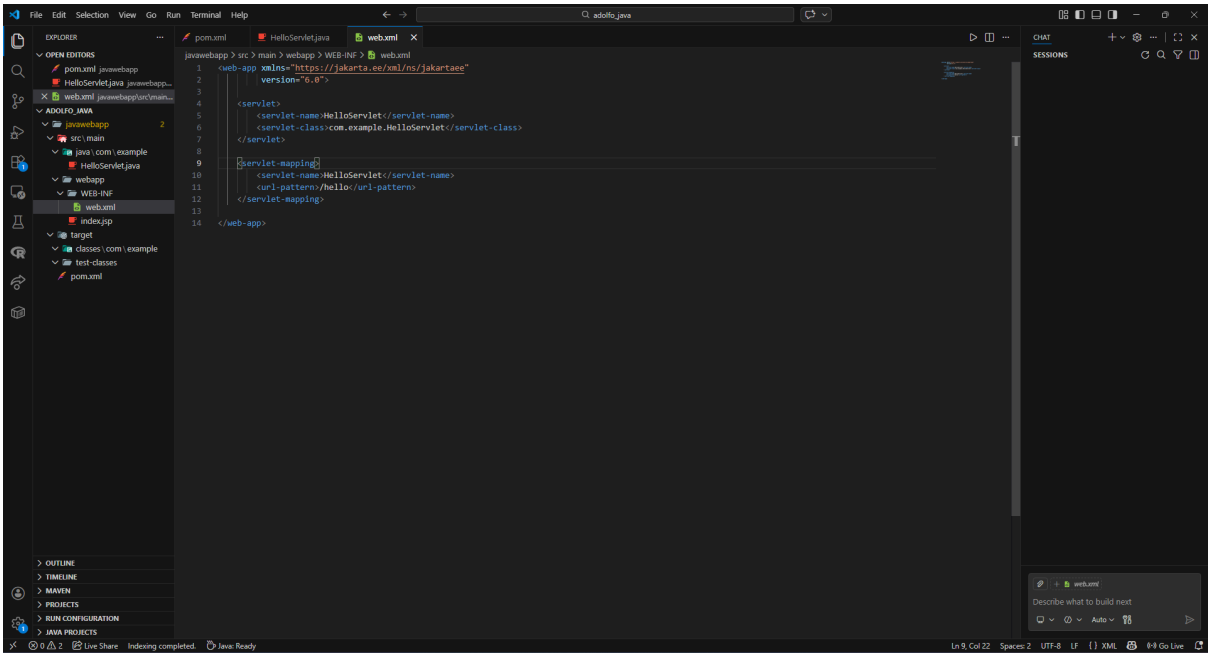
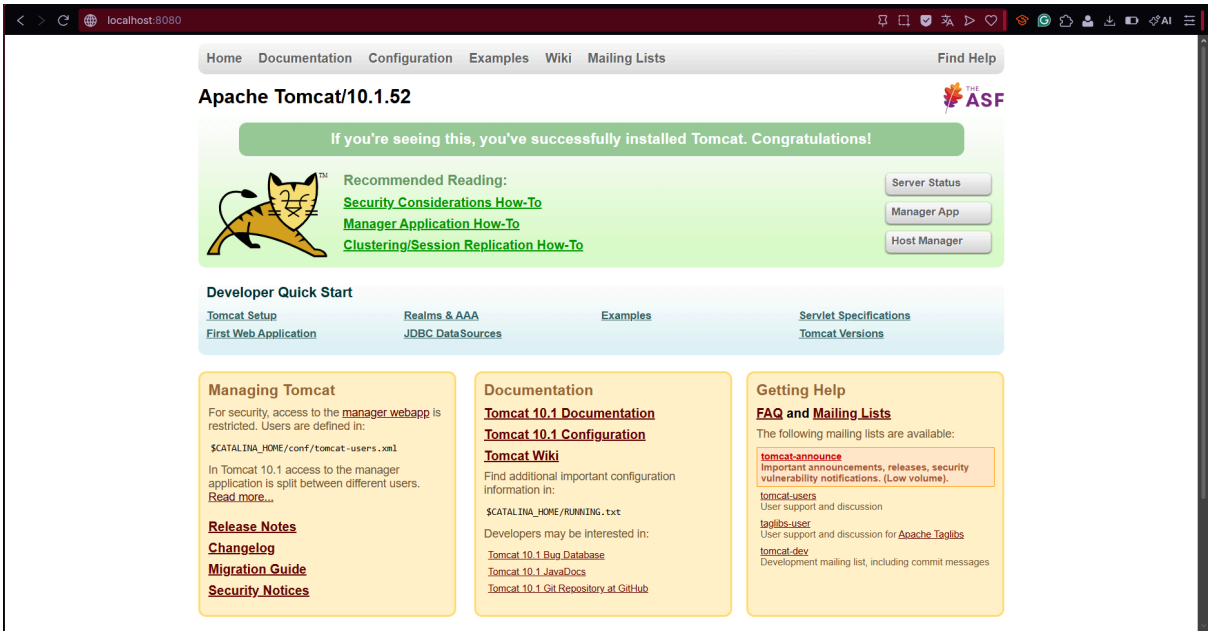
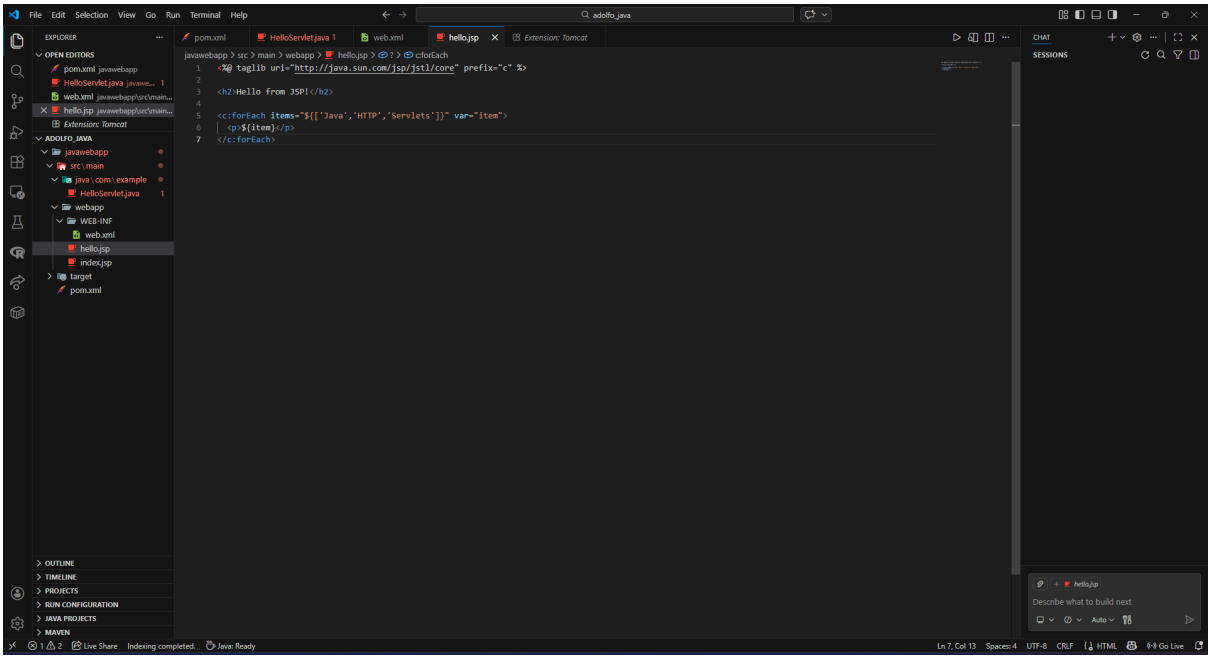
For this evidence, I took some screenshotsto show the access to my active Java web application running locally. I installed the Java Development Kit, configured Maven, and prepared Apache Tomcat for manual deployment. The project includes a Servlet that follows the complete lifecycle defined by the Servlet specification, including initialization, request handling, and termination phases. Through this implementation, the application was packaged and deployed successfully on Tomcat, confirming proper configuration and execution. Additionally, I developed a JSP page enhanced with JSTL tags to render dynamic content. The interaction between the Servlet and JSP components illustrates the roles of controller and view within a layered web architecture. I also implemented a small Spring Boot application. By using dependency injection through the framework's built in container, I was able to observe how the Inversion of Control principle reduces manual configuration and simplifies component management. This comparison highlights the evolution from classic Servlet-based applications to contemporary Spring-based solutions.

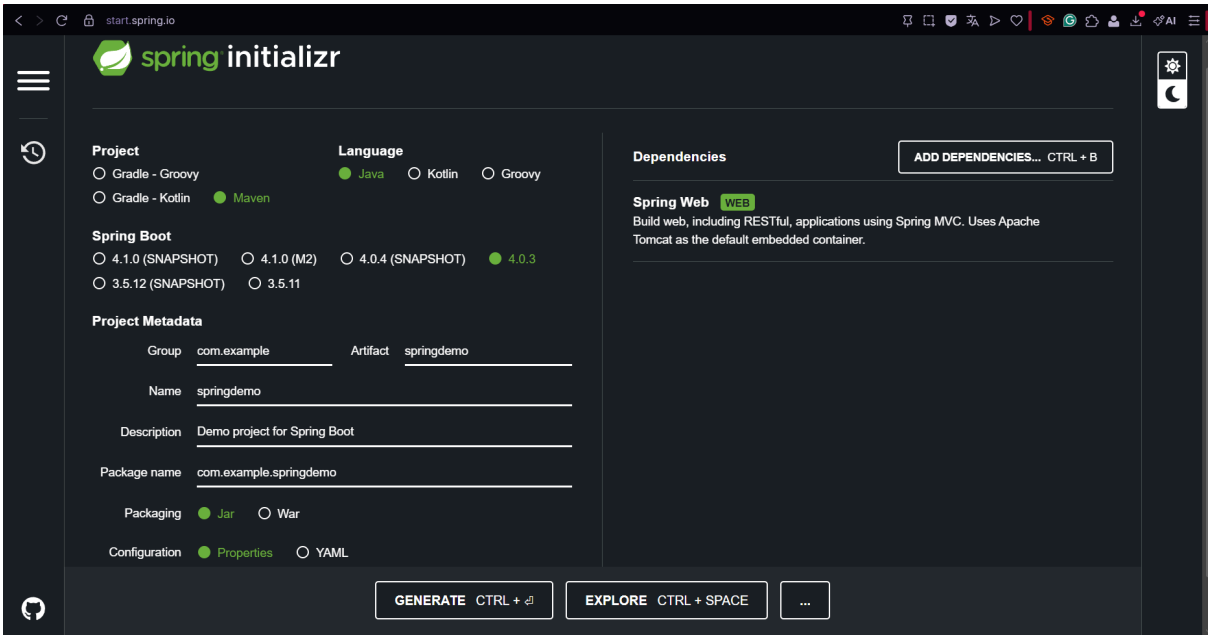


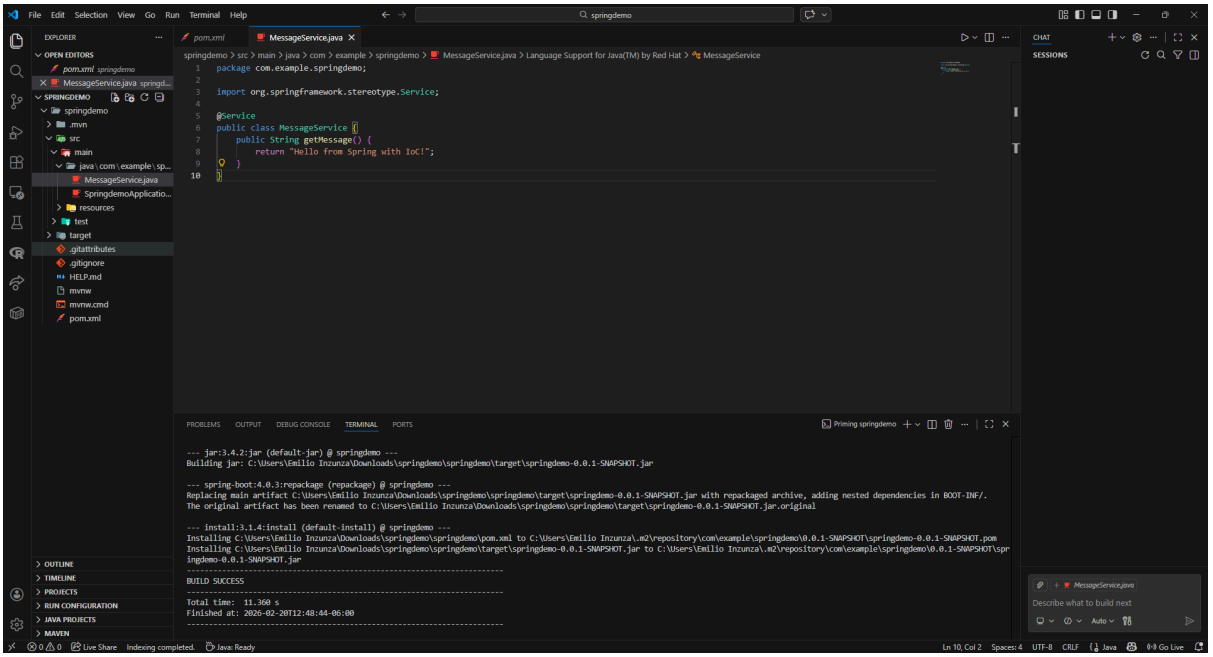

File Edit Selection View Go Run Terminal Help
adolfo_java
EXPLORER
OPEN EDITORS
pom.xml javawebapp
HelloServlet.java javawebapp...
web.xml javawebapp\src\main...
ADOLFO_JAVA
javawebapp 2
src\main
java\com\example
HelloServlet.java
webapp
WEB-INF
web.xml
index.jsp
target
classes\com\example
test-classes
pom.xml
pom.xml
HelloServlet.java
web.xml
javawebapp > src > main > webapp > WEB-INF > web.xml
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
version="6.0">
<servlet>
<servlet-name>HelloServlet</servlet-name>
<servlet-class>com.example.HelloServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>HelloServlet</servlet-name>
<url-pattern>/hello</url-pattern>
</servlet-mapping>
</web-app>
OUTLINE
TIMELINE
MAVEN
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
CHAT
SESSIONS
Describe what to build next
Live Share
Indexing completed.
Java: Ready
Ln 9, Col 22
Spaces: 2
UTF-8
LF
XML
Go Live

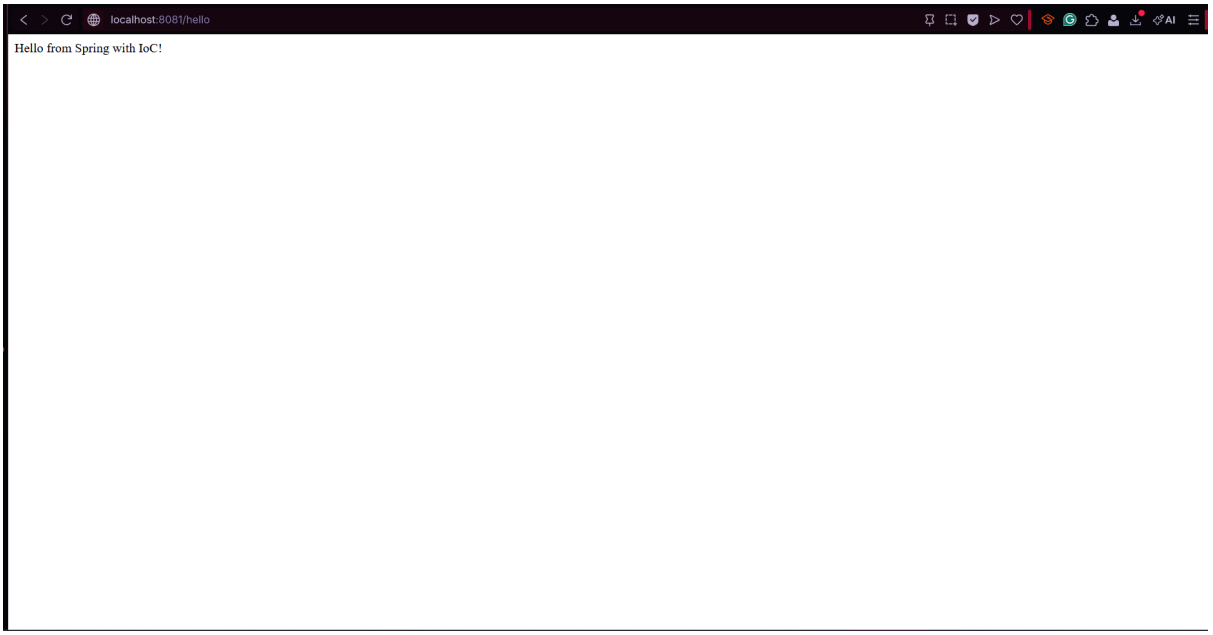

localhost:8080
Home
Documentation
Configuration
Examples
Wiki
Mailing Lists
Find Help
Apache Tomcat/10.1.52
THE ASF
If you're seeing this, you've successfully installed Tomcat. Congratulations!
Recommended Reading:
Security Considerations How-To
Manager Application How-To
Clustering/Session Replication How-To
Server Status
Manager App
Host Manager
Developer Quick Start
Tomcat Setup
First Web Application
Realms & AAA
JDBC DataSources
Examples
Servlet Specifications
Tomcat Versions
Managing Tomcat
For security, access to the manager webapp is restricted. Users are defined in:
$CATALINA_HOME/conf/tomcat-users.xml
In Tomcat 10.1 access to the manager application is split between different users.
Read more...
Release Notes
Changelog
Migration Guide
Security Notices
Documentation
Tomcat 10.1 Documentation
Tomcat 10.1 Configuration
Tomcat Wiki
Find additional important configuration information in:
$CATALINA_HOME/RUNNING.txt
Developers may be interested in:
Tomcat 10.1 Bug Database
Tomcat 10.1 JavaDocs
Tomcat 10.1 Git Repository at GitHub
Getting Help
FAQ and Mailing Lists
The following mailing lists are available:
tomcat-announce
Important announcements, releases, security vulnerability notifications. (Low volume).
tomcat-users
User support and discussion
taglibs-user
User support and discussion for Apache Taglibs
tomcat-dev
Development mailing list, including commit messages


File Edit Selection View Go Run Terminal Help
adolfo_java
EXPLORER
OPEN EDITORS
pom.xml javawebapp
HelloServlet.java javawe... 1
web.xml javawebapp\src\main...
hello.jsp javawebapp\src\main...
Extension: Tomcat
ADOLFO_JAVA
javawebapp
src\main
java\com\example
HelloServlet.java 1
webapp
WEB-INF
web.xml
hello.jsp
index.jsp
target
pom.xml
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
MAVEN
pom.xml
HelloServlet.java 1
web.xml
hello.jsp
Extension: Tomcat
javawebapp > src > main > webapp > hello.jsp > ? > c:forEach
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
<h2>Hello from JSP!</h2>
<c:forEach items="${['Java','HTTP','Servlets']}" var="item">
${item}
</c:forEach>
CHAT
SESSIONS
hello.jsp
Describe what to build next
Auto
Live Share
Indexing completed.
Java: Ready
Ln 7, Col 13
Spaces: 4
UTF-8
CRLF
HTML
Go Live

localhost:8080/javawebapp/hello.jsp
Hello from JSP!
Java
HTTP
Servlets


start.spring.io
spring initializr
Project
Gradle - Groovy
Gradle - Kotlin
Maven
Language
Java
Kotlin
Groovy
Spring Boot
4.1.0 (SNAPSHOT)
4.1.0 (M2)
4.0.4 (SNAPSHOT)
4.0.3
3.5.12 (SNAPSHOT)
3.5.11
Project Metadata
Group
com.example
Artifact
springdemo
Name
springdemo
Description
Demo project for Spring Boot
Package name
com.example.springdemo
Packaging
Jar
War
Configuration
Properties
YAML
Dependencies
ADD DEPENDENCIES... CTRL + B
Spring Web
WEB
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.
GENERATE CTRL + ⏎
EXPLORE CTRL + SPACE
...


File Edit Selection View Go Run Terminal Help
springdemo
EXPLORER
OPEN EDITORS
pom.xml springdemo
MessageService.java springd...
SPRINGDEMO
springdemo
.mvn
src
main
java \ com \ example \ sp...
MessageService.java
SpringdemoApplicatio...
resources
test
target
.gitattributes
.gitignore
HELP.md
mvnw
mvnw.cmd
pom.xml
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
MAVEN
pom.xml
MessageService.java
springdemo > src > main > java > com > example > springdemo > MessageService.java > Language Support for Java(TM) by Red Hat > MessageService
package com.example.springdemo;
import org.springframework.stereotype.Service;
@Service
public class MessageService {
public String getMessage() {
return "Hello from Spring with IoC!";
}
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Priming springdemo
--- jar:3.4.2:jar (default-jar) @ springdemo ---
Building jar: C:\Users\Emilio Inzunza\Downloads\springdemo\springdemo\target\springdemo-0.0.1-SNAPSHOT.jar
--- spring-boot:4.0.3:repackage (repackage) @ springdemo ---
Replacing main artifact C:\Users\Emilio Inzunza\Downloads\springdemo\springdemo\target\springdemo-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
The original artifact has been renamed to C:\Users\Emilio Inzunza\Downloads\springdemo\springdemo\target\springdemo-0.0.1-SNAPSHOT.jar.original
--- install:3.1.4:install (default-install) @ springdemo ---
Installing C:\Users\Emilio Inzunza\Downloads\springdemo\springdemo\pom.xml to C:\Users\Emilio Inzunza\.m2\repository\com\example\springdemo\0.0.1-SNAPSHOT\springdemo-0.0.1-SNAPSHOT.pom
Installing C:\Users\Emilio Inzunza\Downloads\springdemo\springdemo\target\springdemo-0.0.1-SNAPSHOT.jar to C:\Users\Emilio Inzunza\.m2\repository\com\example\springdemo\0.0.1-SNAPSHOT\springdemo-0.0.1-SNAPSHOT.jar
BUILD SUCCESS
Total time: 11.360 s
Finished at: 2026-02-20T12:48:44-06:00
CHAT
SESSIONS
MessageService.java
Describe what to build next
Auto
Live Share
Indexing completed.
Java: Ready
Ln 10, Col 2
Spaces: 4
UTF-8
CRLF
Java
Go Live



Running Application Endpoints

The following URLs correspond to the running applications demonstrated in this exercise:

- Servlet (Tomcat deployment): http://localhost:8080/javawebapp/hello
- JSP with JSTL: http://localhost:8080/javawebapp/hello.jsp
- Spring Boot Service (IoC): http://localhost:8081/hello